

Digital SAT Practice Test 10 — Module 1

Math · Full Walkthrough

All 27 Math questions transcribed verbatim, with the College Board answer key, a step-by-step worked solution for every item, and a plain-English explanation for every correct and incorrect choice. Built by SOMATH — the math school on the Upper West Side of NYC — for students preparing for the Digital SAT.

How to use this packet

Work each question with paper and pencil before you look at the solution. Cover the answer, read the prompt twice, set up the equation or figure on your own, then check. You are not trying to memorize solutions — you are trying to build the reflex of recognizing which SAT-Math tool the question wants (Pythagorean theorem, exponent rules, substitution, sum of roots, etc.) before you touch the calculator.

What Math Module 1 tests

The Digital SAT Math section has two 35-minute modules of 22 questions each (20 scored + 2 pretest). Module 1 is not adaptive — every student sees the same questions. Your score on Module 1 determines which version of Module 2 you get. The Bluebook calculator (Desmos) is available for every question. The four domains tested (weighted roughly 35% / 35% / 15% / 15% across the section) are:

Algebra. Linear equations & inequalities in one and two variables, systems of linear equations, and interpreting linear models.

Advanced Math. Quadratic and exponential functions, function notation and transformations, and equivalent expressions.

Problem-Solving & Data Analysis. Ratios, percents, rates, two-way tables, probability, and modeling from data.

Geometry & Trigonometry. Area & volume, triangle congruence and similarity, right-triangle trigonometry, and circles in the coordinate plane.

Practice Questions

Work these first. Full solutions start on the solutions pages.

Section 1 · Algebra & Linear Systems

Linear equations & inequalities, linear systems, and function evaluation — the fastest-paced part of the module.

Question 1 · Reading a Line Graph

[Figure: Line graph of percent of cars for sale by model year 2010 to 2019. Values (approx): 2010 12%, 2011 12%, 2012 12%, 2013 10%, 2014 4% (minimum), 2015 10%, 2016 10%, 2017 11%, 2018 11%, 2019 11%.]

The line graph shows the percent of cars for sale at a used car lot on a given day by model year. For what model year is the percent of cars for sale the smallest?

- A) 2012
- B) 2013
- C) 2014
- D) 2015

Question 2 · System of Linear Equations (Graphical)

[Figure: xy -coordinate plane showing two lines intersecting at $(0, 3)$. One line passes through $(0, 3)$ with negative slope, the other line also passes through $(0, 3)$ with a steeper negative slope.]

The graph of a system of linear equations is shown. What is the solution (x, y) to the system?

- A) $(4, -5)$
- B) $(0, 3)$
- C) $(0, -2)$
- D) $(-2, 3)$

Question 3 · Writing a Linear Inequality

The total cost, in dollars, to rent a surfboard consists of a \$25 service fee and a \$10 per hour rental fee. A person rents a surfboard for t hours and intends to spend a maximum of \$75 to rent the surfboard. Which inequality represents this situation?

- A) $10t \leq 75$
- B) $10 + 25t \leq 75$
- C) $25t \leq 75$
- D) $25 + 10t \leq 75$

Question 4 · Translating a Graph (Function Transformation)

[Figure: Parabola opening upward with vertex near $(3, -2)$, passing through the x -axis at roughly $x=1$ and $x=5$.]

The graph shown will be translated up 4 units. Which of the following will be the resulting graph?

- A) Parabola with vertex at $(3, 2)$, opening upward, passing through the x -axis above $y = 2$.
- B) Parabola with vertex at $(3, -6)$, opening upward.
- C) Parabola with vertex at $(-3, -2)$, opening upward.
- D) Parabola with vertex at $(7, -2)$, opening upward.

Question 5 · Evaluating a Linear Function

$s = 40 + 3t$. The equation gives the speed s , in miles per hour, of a certain car t seconds after it began to accelerate. What is the speed, in miles per hour, of the car 5 seconds after it began to accelerate?

- A) 40
- B) 43
- C) 45
- D) 55

Question 6 · Evaluating a Quadratic Function

The function f is defined by $f(x) = x^2 + x + 71$. What is the value of $f(2)$?

Student-produced response — enter your answer as a fraction or decimal.

Question 7 · Linear Inequality (Budget Constraint)

An event planner is planning a party. It costs the event planner a one-time fee of \$35 to rent the venue and \$10.25 per attendee. The event planner has a budget of \$300. What is the greatest number of attendees possible without exceeding the budget?

Student-produced response — enter your answer as a fraction or decimal.

Question 8 · Two-Way Table Probability

[The question includes a data table. See the online post for the full table.]

The table gives the distribution of votes for a new school mascot and grade level for 80 students. If one of these students is selected at random, what is the probability of selecting a student whose vote for new mascot was for a lion?

- A) $\frac{1}{9}$
- B) $\frac{1}{5}$
- C) $\frac{1}{4}$
- D) $\frac{2}{3}$

Question 9 · Congruent Triangles (Angle Sum)

Triangles ABC and DEF are congruent, where A corresponds to D, and B and E are right angles. The measure of angle A is 18° . What is the measure of angle F?

- A) 18°
- B) 72°
- C) 90°
- D) 162°

Question 10 · Scaling a Linear Equation

If $4x + 2 = 12$, what is the value of $16x + 8$?

- A) 40
- B) 48
- C) 56
- D) 60

Question 11 · Exponent Rules (Product of Monomials)

Which expression is equivalent to $(m^4q^4z^{-1})(mq^5z^3)$, where m, q, and z are positive?

- A) $m^4q^{20}z^{-3}$
- B) $m^5q^9z^2$
- C) $m^6q^8z^{-1}$
- D) $m^{20}q^{12}z^{-2}$

Question 12 · Modeling with Functions (Linear vs Exponential)

An airplane descends from an altitude of 9,500 feet to 5,000 feet at a constant rate of 400 feet per minute. What type of function best models the relationship between the descending airplane's altitude and time?

- A) Decreasing exponential
- B) Decreasing linear
- C) Increasing exponential
- D) Increasing linear

Question 13 · Solving a Linear System

$3x + 6 = 4y$ and $3x + 4 = 2y$. The solution to the given system of equations is (x, y) . What is the value of y?

Student-produced response — enter your answer as a fraction or decimal.

Section 2 · Advanced Math & Functions

Quadratics, exponentials, function transformations, and interpreting function notation in context.

Question 14 · Function Transformation (Vertical Shift)

The function f is defined by $f(x) = (x - 6)(x - 2)(x + 6)$. In the xy -plane, the graph of $y = g(x)$ is the result of translating the graph of $y = f(x)$ up 4 units. What is the value of $g(0)$?

Student-produced response — enter your answer as a fraction or decimal.

Question 15 · Interpreting Function Notation in Context

The function $f(w) = 6w^2$ gives the area of a rectangle, in square feet (ft^2), if its width is w ft and its length is 6 times its width. Which of the following is the best interpretation of $f(14) = 1,176$?

- A) If the width of the rectangle is 14 ft, then the area of the rectangle is $1,176 \text{ ft}^2$.
- B) If the width of the rectangle is 14 ft, then the length of the rectangle is 1,176 ft.
- C) If the width of the rectangle is 1,176 ft, then the length of the rectangle is 14 ft.
- D) If the width of the rectangle is 1,176 ft, then the area of the rectangle is 14 ft^2 .

Question 16 · Exponential Growth Model (Doubling)

The number of bacteria in a liquid medium doubles every day. There are 44,000 bacteria in the liquid medium at the start of an observation. Which of the following represents the number of bacteria, y , in the liquid medium t days after the start of the observation?

- A) $y = (1/2)(44,000)^t$
- B) $y = 2(44,000)^t$
- C) $y = 44,000(1/2)^t$
- D) $y = 44,000(2)^t$

Question 17 · Exponential Function from a Table

[The question includes a data table. See the online post for the full table.]

The table shows the exponential relationship between the number of years, x , since Hana started training in pole vault, and the estimated height $h(x)$, in meters, of her best pole vault for that year. Which of the following functions best represents this relationship, where $x \leq 4$?

- A) $h(x) = 1.12(0.23)^x$
- B) $h(x) = 1.12(1.23)^x$
- C) $h(x) = 1.23(0.12)^x$
- D) $h(x) = 1.23(1.12)^x$

Question 18 • Intercepts of a Linear Function

The function h is defined by $h(x) = 4x + 28$. The graph of $y = h(x)$ in the xy -plane has an x -intercept at $(a, 0)$ and a y -intercept at $(0, b)$, where a and b are constants. What is the value of $a + b$?

- A) 21
- B) 28
- C) 32
- D) 35

Question 19 • Linear Inequality — Table Check

[The question includes a data table. See the online post for the full table.]

$y < 5x + 6$. For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

- A) Table A: (3, 17), (5, 27), (7, 37).
- B) Table B: (3, 17), (5, 35), (7, 37).
- C) Table C: (3, 25), (5, 35), (7, 45).
- D) Table D: (3, 21), (5, 31), (7, 41).

Question 20 • Solving a Linear System

$y = 4x + 1$ and $4y = 15x - 8$. The solution to the given system of equations is (x, y) . What is the value of $x - y$?

Student-produced response — enter your answer as a fraction or decimal.

Section 3 • Geometry & Trigonometry

Pythagorean theorem, scale factors and area, equations of circles, and triangle congruence.

Question 21 • Pythagorean Theorem (Simplifying a Radical)

A right triangle has legs with lengths of 24 centimeters and 21 centimeters. If the length of this triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$, where d is an integer, what is the value of d ?

Student-produced response — enter your answer as a fraction or decimal.

Question 22 • Scale Factor and Area

The floor of a ballroom has an area of 600 square meters. An architect creates a scale model of the floor of the ballroom, where the length of each side of the model is $\frac{1}{10}$ times the length of the corresponding side of the actual floor of the ballroom. What is the area, in square meters, of the scale model?

- A) 6
- B) 10
- C) 60
- D) 150

Question 23 · Equation of a Circle (Tangent to Axis)

Which of the following equations represents a circle in the xy -plane that intersects the y -axis at exactly one point?

- A) $(x - 8)^2 + (y - 8)^2 = 16$
- B) $(x - 8)^2 + (y - 4)^2 = 16$
- C) $(x - 4)^2 + (y - 9)^2 = 16$
- D) $x^2 + (y - 9)^2 = 16$

Question 24 · Triangle Congruence (SSS/SAS/ASA/AAS)

In triangles ABC and DEF, angles B and E each have measure 27° and angles C and F each have measure 41° . Which additional piece of information is sufficient to determine whether triangle ABC is congruent to triangle DEF?

- A) The measure of angle A
- B) The length of side AB
- C) The lengths of sides BC and EF
- D) No additional information is necessary.

Section 4 · Problem-Solving, Data Analysis & Advanced Algebra

Percent change, piecewise cost models, and the sum of roots of a quadratic — the last three items on the module.

Question 25 · Percent Increase (Solving Backwards)

The result of increasing the quantity x by 1,800% is 684. What is the value of x ?

- A) 12,996
- B) 12,312
- C) 38
- D) 36

Question 26 · Piecewise Linear Cost Function

A window repair specialist charges \$220 for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 5 hours of repair is \$400. Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

- A) $f(x) = 60x + 100$
- B) $f(x) = 60x + 220$
- C) $f(x) = 80x$
- D) $f(x) = 80x + 220$

Question 27 · Sum of Roots (Quadratic)

$x(x + 1) - 56 = 4x(x - 7)$. What is the sum of the solutions to the given equation?

Student-produced response — enter your answer as a fraction or decimal.

Solutions & Explanations

Correct answer, worked solution, and a plain-English explanation for every choice.

Question 1 · Reading a Line Graph

Correct answer: C) 2014

Worked solution

Read the y -value of each labeled dot on the line graph and pick the model year with the lowest point.

2010: $\approx 12\%$

2011: $\approx 12\%$

2012: $\approx 12\%$

2013: $\approx 10\%$

2014: $\approx 4\%$ ← minimum

2015: $\approx 10\%$

× A) 2012

2012 sits at about 12%, one of the highest points on the graph, not the lowest.

× B) 2013

2013 sits at about 10% — lower than 2010–2012 but still well above 2014.

✓ C) 2014

2014 is the single lowest dot on the graph, at about 4%. This is the model year with the smallest percent of cars for sale.

× D) 2015

2015 sits at about 10%, back above the 2014 low.

Question 2 · System of Linear Equations (Graphical)

Correct answer: A) (4, -5)

Worked solution

The solution to a system shown on a graph is the point where the two lines intersect.

Read the intersection point from the graph: $(x, y) = (4, -5)$.

✓ A) (4, -5)

The two lines cross at the point four units right of the origin and five units down. That is (4, -5).

× B) (0, 3)

(0, 3) is the y -intercept of one line, not the intersection of both lines.

× C) (0, -2)

(0, -2) is not a point where either line crosses an axis; not the intersection.

× D) (-2, 3)

(-2, 3) is not the intersection point of the two lines.

Question 3 · Writing a Linear Inequality

Correct answer: D) $25 + 10t \leq 75$

Worked solution

Total cost = fixed service fee + (hourly fee × hours).

Total cost = $25 + 10t$.

Maximum \$75 → $25 + 10t \leq 75$.

× A) $10t \leq 75$

Ignores the \$25 service fee entirely.

× B) $10 + 25t \leq 75$

Reverses which number is the fixed fee and which is the hourly rate. \$10 is per hour, not the fixed fee.

× C) $25t \leq 75$

Treats \$25 as the hourly rate and ignores the \$10 hourly rate — reversed.

✓ D) $25 + 10t \leq 75$

The \$25 service fee is fixed, \$10 is per hour for t hours ($10t$), and the total must be at most \$75, so $25 + 10t \leq 75$.

Question 4 · Translating a Graph (Function Transformation)

Correct answer: A) Parabola with vertex at $(3, 2)$, opening upward, passing through the x -axis above $y = 2$.

Worked solution

Translating a graph up by 4 units means every point (x, y) on the original graph moves to $(x, y + 4)$.

The original vertex is at about $(3, -2)$.

New vertex: $(3, -2 + 4) = (3, 2)$.

The shape does not change — only the height.

✓ A) Parabola with vertex at $(3, 2)$, opening upward, passing through the x -axis above $y = 2$.

The vertex moves from $(3, -2)$ up to $(3, 2)$. Every point on the parabola shifts up by 4 units. This matches choice A.

× B) Parabola with vertex at $(3, -6)$, opening upward.

This would be a shift down by 4, not up.

× C) Parabola with vertex at $(-3, -2)$, opening upward.

This shifts the parabola left, not up.

× D) Parabola with vertex at $(7, -2)$, opening upward.

This shifts the parabola right, not up.

Question 5 · Evaluating a Linear Function

Correct answer: D) 55

Worked solution

Substitute $t = 5$ into $s = 40 + 3t$.

$$s = 40 + 3(5) = 40 + 15 = 55.$$

× A) 40

This is the starting speed at $t = 0$. The question asks for speed at $t = 5$ seconds.

× B) 43

This is $40 + 3$ — it uses $t = 1$, not $t = 5$.

× C) 45

This is $40 + 5$ — it forgets to multiply 5 by the coefficient 3.

✓ D) 55

Substituting $t = 5$ gives $s = 40 + 3(5) = 40 + 15 = 55$ mph.

Question 6 · Evaluating a Quadratic Function

Correct answer: 77

Worked solution

Substitute $x = 2$ into $f(x) = x^2 + x + 71$.

$$f(2) = 2^2 + 2 + 71$$

$$= 4 + 2 + 71$$

$$= 77.$$

Why this works

Substituting $x = 2$ gives $f(2) = 4 + 2 + 71 = 77$.

Question 7 · Linear Inequality (Budget Constraint)

Correct answer: 25

Worked solution

Let n = number of attendees.

$$\text{Total cost: } 35 + 10.25n \leq 300$$

$$10.25n \leq 265$$

$$n \leq 265 / 10.25 \approx 25.85.$$

n must be a whole number of attendees, so $n = 25$.

Why this works

The budget inequality $35 + 10.25n \leq 300$ gives $n \leq 25.85$. Since the number of attendees must be a whole number, the greatest possible value is 25.

Question 8 · Two-Way Table Probability

Correct answer: C) $\frac{1}{4}$

Worked solution

$$\begin{aligned} P(\text{Lion}) &= (\text{Total votes for Lion}) / (\text{Total students}) \\ &= 20 / 80 \\ &= \frac{1}{4}. \end{aligned}$$

× A) $\frac{1}{9}$

$\frac{1}{9}$ is not the ratio $20/80$. This does not match any row total.

× B) $\frac{1}{5}$

$\frac{1}{5} = 16/80$. There are 20 lion votes, not 16.

✓ C) $\frac{1}{4}$

20 students voted Lion out of 80 total $\rightarrow 20/80 = \frac{1}{4}$.

× D) $\frac{2}{3}$

$\frac{2}{3} \approx 53/80$ — far more than any single mascot received.

Question 9 · Congruent Triangles (Angle Sum)

Correct answer: B) 72°

Worked solution

Corresponding angles of congruent triangles are equal, so angle F corresponds to angle C.

Angles in a triangle sum to 180° .

In triangle ABC: $A + B + C = 180^\circ$.

$18^\circ + 90^\circ + C = 180^\circ \rightarrow C = 72^\circ$.

So angle F = 72° .

× A) 18°

18° is the measure of angle A, which corresponds to angle D — not angle F.

✓ B) 72°

Angle F corresponds to angle C. Since $A + B + C = 180^\circ$ and $A = 18^\circ$, $B = 90^\circ$, we get $C = 72^\circ = F$.

× C) 90°

90° is the measure of angle B (and its corresponding angle E), not angle F.

× D) 162°

$162^\circ = 180^\circ - 18^\circ$. This ignores the right angle at B.

Question 10 · Scaling a Linear Equation

Correct answer: B) 48

Worked solution

Multiply both sides of $4x + 2 = 12$ by 4:

$$4(4x + 2) = 4(12)$$

$$16x + 8 = 48.$$

(No need to solve for x .)

× A) 40

40 comes from solving $4x = 10 \rightarrow x = 2.5$, then $16(2.5) = 40$ — but this forgets the +8 term.

✓ B) 48

$16x + 8$ is exactly 4 times $4x + 2$, so it equals $4(12) = 48$.

× C) 56

56 would require $4x + 2 = 14$, not 12.

× D) 60

$60 = 5 \times 12$, but $16x + 8$ is $4 \times (4x + 2)$, not $5 \times$.

Question 11 · Exponent Rules (Product of Monomials)

Correct answer: B) $m^5q^9z^2$

Worked solution

When multiplying like bases, add the exponents.

$$m: 4 + 1 = 5$$

$$q: 4 + 5 = 9$$

$$z: -1 + 3 = 2$$

Result: $m^5q^9z^2$.

× A) $m^4q^{20}z^{-3}$

This multiplies the q exponents ($4 \times 5 = 20$) instead of adding them, and mishandles the m and z exponents too.

✓ B) $m^5q^9z^2$

Add exponents of like bases: $m^{(4+1)} = m^5$, $q^{(4+5)} = q^9$, $z^{(-1+3)} = z^2$. This gives $m^5q^9z^2$.

× C) $m^6q^8z^{-1}$

$6 = 4 + 1 + 1$ (extra m) and $8 = 4 + 4$ (wrong for q). Doesn't match adding once correctly.

× D) $m^{20}q^{12}z^{-2}$

Multiplies the m exponents ($4 \times 1 \times 5 = 20$) — exponents add when multiplying like bases, they don't multiply.

Question 12 · Modeling with Functions (Linear vs Exponential)

Correct answer: B) Decreasing linear

Worked solution

"Constant rate" of change = linear function (same change per unit time).

"Descending" means altitude is going down as time increases → decreasing.

Combined: decreasing linear.

× A) Decreasing exponential

Exponential decay changes by a constant percent, not a constant amount, per unit time. The airplane loses 400 ft each minute (a constant amount), not a constant percent.

✓ B) Decreasing linear

A constant rate of change (400 ft/min every minute) is the signature of a linear function. Since altitude is decreasing, the linear function is decreasing.

× C) Increasing exponential

Altitude is decreasing, not increasing.

× D) Increasing linear

Altitude is decreasing, not increasing.

Question 13 · Solving a Linear System

Correct answer: 1

Worked solution

Both equations start with $3x$. Subtract the second from the first:

$$(3x + 6) - (3x + 4) = 4y - 2y$$

$$2 = 2y$$

$$y = 1.$$

Why this works

Subtracting the second equation from the first eliminates x and gives $2 = 2y$, so $y = 1$.

Question 14 · Function Transformation (Vertical Shift)

Correct answer: 76

Worked solution

A shift up by 4 units means $g(x) = f(x) + 4$.

First find $f(0)$:

$$f(0) = (0 - 6)(0 - 2)(0 + 6)$$

$$= (-6)(-2)(6)$$

$$= 72.$$

$$\text{Then } g(0) = f(0) + 4 = 72 + 4 = 76.$$

Why this works

Translating up 4 units means $g(x) = f(x) + 4$. $f(0) = (-6)(-2)(6) = 72$, so $g(0) = 72 + 4 = 76$.

Question 15 · Interpreting Function Notation in Context

Correct answer: A) If the width of the rectangle is 14 ft, then the area of the rectangle is 1,176 ft².

Worked solution

In $f(w) = 6w^2$, the input w is the width (in ft) and the output $f(w)$ is the area (in ft²).

So $f(14) = 1,176$ means: when the width is 14 ft, the area is 1,176 ft².

✓ A) If the width of the rectangle is 14 ft, then the area of the rectangle is 1,176 ft².

The input 14 represents width in ft, and the output 1,176 represents area in ft². So $f(14) = 1,176$ means width 14 ft gives area 1,176 ft².

✗ B) If the width of the rectangle is 14 ft, then the length of the rectangle is 1,176 ft.

1,176 is the area, not the length. The length would be $6 \cdot 14 = 84$ ft.

✗ C) If the width of the rectangle is 1,176 ft, then the length of the rectangle is 14 ft.

Reverses input and output: 14 is the input (width), not the output.

✗ D) If the width of the rectangle is 1,176 ft, then the area of the rectangle is 14 ft².

Reverses input and output and swaps units.

Question 16 · Exponential Growth Model (Doubling)

Correct answer: D) $y = 44,000(2)^t$

Worked solution

Exponential growth: $y = (\text{starting value}) \cdot (\text{growth factor})^t$.

Starting value = 44,000.

Doubles each day → growth factor = 2.

$y = 44,000 \cdot (2)^t$.

✗ A) $y = (1/2)(44,000)^t$

The starting value should be the base 44,000, not (1/2). Also, doubling means multiply by 2, not by (1/2).

✗ B) $y = 2(44,000)^t$

Uses 44,000 as the base of the exponent, which grows far too fast.

✗ C) $y = 44,000(1/2)^t$

$(1/2)^t$ means the bacteria are halving each day, but the problem says doubling.

✓ D) $y = 44,000(2)^t$

Starting with 44,000 and doubling each day gives $y = 44,000 \cdot (2)^t$.

Question 17 · Exponential Function from a TableCorrect answer: D) $h(x) = 1.23(1.12)^x$

Worked solution

Exponential model: $h(x) = a \cdot b^x$.**At $x = 0$:** $h(0) = a \cdot b^0 = a$. Table says $h(0) = 1.23$, so $a = 1.23$.**Find growth factor b using $x = 2$:** $h(2) = 1.23 \cdot b^2 = 1.54$. $b^2 = 1.54 / 1.23 \approx 1.2520$. $b \approx \sqrt{1.2520} \approx 1.12$.**So $h(x) = 1.23(1.12)^x$.** Check $x = 4$: $1.23 \cdot (1.12)^4 \approx 1.23 \cdot 1.5735 \approx 1.94 \checkmark$.× A) $h(x) = 1.12(0.23)^x$ **$a = 1.12$ is wrong:** $h(0)$ in the table is 1.23, not 1.12. Base 0.23 would also decrease h , not increase it.× B) $h(x) = 1.12(1.23)^x$ **Correct base (1.23) but wrong a :** 1.12 doesn't match $h(0) = 1.23$.× C) $h(x) = 1.23(0.12)^x$ **Base 0.12 causes h to shrink rapidly,** contradicting the increasing values in the table.✓ D) $h(x) = 1.23(1.12)^x$ **$a = h(0) = 1.23$, and multiplying by 1.12 each year matches:** $1.23 \rightarrow 1.54 \rightarrow 1.94$, consistent with the table.**Question 18 · Intercepts of a Linear Function**

Correct answer: A) 21

Worked solution

y-intercept: set $x = 0 \rightarrow h(0) = 4(0) + 28 = 28$. So $b = 28$.**x-intercept:** set $h(x) = 0 \rightarrow 4x + 28 = 0 \rightarrow x = -7$. So $a = -7$. $a + b = -7 + 28 = 21$.

✓ A) 21

 $a = -7$ (from $4x + 28 = 0$) and $b = 28$ (from $x = 0$). Sum: $-7 + 28 = 21$.

× B) 28

28 is just b , the y-intercept — forgets to add a .

× C) 32

32 would require $a = 4$, but $4x + 28 = 0$ gives $x = -7$, not 4.

× D) 35

 $35 = 28 + 7$. Uses $|a|$ instead of a . The x-intercept a is negative, not positive.

Question 19 · Linear Inequality — Table Check

Correct answer: A) Table A: (3, 17), (5, 27), (7, 37).

Worked solution

Compute $5x + 6$ for each row and require $y < 5x + 6$.

A: (3,17): $5(3)+6 = 21$; $17 < 21$ ✓ ; (5,27): 31 ; $27 < 31$ ✓ ; (7,37): 41 ; $37 < 41$ ✓ — all satisfy.

B: (3,17): $17 < 21$ ✓ ; (5,35): $5(5)+6=31$; $35 < 31$? No.

C: (3,25): $25 < 21$? No.

D: (3,21): $21 < 21$? No — the inequality is strict.

✓ A) Table A: (3, 17), (5, 27), (7, 37).

All three rows satisfy $y < 5x + 6$: $17 < 21$, $27 < 31$, $37 < 41$.

✗ B) Table B: (3, 17), (5, 35), (7, 37).

At $x = 5$, $5(5)+6 = 31$, and 35 is NOT less than 31. Fails.

✗ C) Table C: (3, 25), (5, 35), (7, 45).

At $x = 3$, $5(3)+6 = 21$, and 25 is NOT less than 21. Fails at the very first row.

✗ D) Table D: (3, 21), (5, 31), (7, 41).

At $x = 3$, $y = 21$ and $5(3)+6 = 21$. The inequality is strict ($<$), so $21 < 21$ is false.

Question 20 · Solving a Linear System

Correct answer: 35

Worked solution

Substitute $y = 4x + 1$ into $4y = 15x - 8$:

$$4(4x + 1) = 15x - 8$$

$$16x + 4 = 15x - 8$$

$$x = -12.$$

$$\text{Then } y = 4(-12) + 1 = -47.$$

$$x - y = -12 - (-47) = -12 + 47 = 35.$$

Why this works

Substituting $y = 4x + 1$ gives $16x + 4 = 15x - 8$, so $x = -12$ and $y = -47$. Then $x - y = -12 - (-47) = 35$.

Question 21 · Pythagorean Theorem (Simplifying a Radical)

Correct answer: 113

Worked solution

Pythagorean theorem: $c^2 = a^2 + b^2$.

$$c^2 = 24^2 + 21^2 = 576 + 441 = 1,017.$$

$$c = \sqrt{1,017}.$$

Factor out the largest perfect square:

$$1,017 = 9 \cdot 113 \text{ (113 is prime).}$$

$$c = \sqrt{(9 \cdot 113)} = 3\sqrt{113}.$$

So $d = 113$.

Why this works

$$c = \sqrt{(24^2 + 21^2)} = \sqrt{1,017} = \sqrt{(9 \cdot 113)} = 3\sqrt{113}, \text{ so } d = 113.$$

Question 22 · Scale Factor and Area

Correct answer: A) 6

Worked solution

When you scale all lengths by a factor k , the area scales by k^2 .

Here $k = 1/10$, so $k^2 = 1/100$.

Model area = $600 \cdot (1/100) = 6$ square meters.

✓ A) 6

Sides scale by $1/10$, so area scales by $(1/10)^2 = 1/100$. $600 \cdot 1/100 = 6$.

× B) 10

$10 = 600 \cdot (1/60)$. This treats the linear scale factor as an area scale factor incorrectly.

× C) 60

$60 = 600 \cdot (1/10)$. This uses the linear scale factor $1/10$ on the area directly — but area scales by the SQUARE of the linear factor.

× D) 150

$150 = 600 \cdot (1/4)$. This uses $(1/2)^2$ instead of $(1/10)^2$.

Question 23 · Equation of a Circle (Tangent to Axis)

Correct answer: C) $(x - 4)^2 + (y - 9)^2 = 16$

Worked solution

In $(x-h)^2 + (y-k)^2 = r^2$, center = (h, k) and radius = r .

A circle intersects the y -axis at exactly one point when it is TANGENT to the y -axis:

$$|h| = r.$$

All four choices have $r^2 = 16$, so $r = 4$.

We need $|h| = 4$.

A) center $(8, 8)$: $|h| = 8 \neq 4 \rightarrow$ does not touch the y -axis at all.

B) center $(8, 4)$: $|h| = 8 \neq 4 \rightarrow$ does not touch the y -axis at all.

C) center $(4, 9)$: $|h| = 4 = r \rightarrow$ tangent to the y -axis at exactly one point ✓.

D) center $(0, 9)$: $|h| = 0 < 4 \rightarrow$ crosses the y -axis at two points.

× A) $(x - 8)^2 + (y - 8)^2 = 16$

Center $(8, 8)$, radius 4. Distance from center to the y -axis is 8, which is greater than 4, so the circle does not touch the y -axis at all.

× B) $(x - 8)^2 + (y - 4)^2 = 16$

Center $(8, 4)$, radius 4. Distance from center to the y -axis is 8, greater than 4, so it does not touch the y -axis.

✓ C) $(x - 4)^2 + (y - 9)^2 = 16$

Center $(4, 9)$, radius 4. The distance from the center to the y -axis equals the radius, so the circle is tangent to the y -axis — it touches at exactly one point.

× D) $x^2 + (y - 9)^2 = 16$

Center $(0, 9)$, radius 4. The center is ON the y -axis, so the circle crosses the y -axis at two points, not one.

Question 24 · Triangle Congruence (SSS/SAS/ASA/AAS)

Correct answer: C) The lengths of sides BC and EF

Worked solution

We already know two pairs of corresponding angles are equal ($B = E = 27^\circ$, $C = F = 41^\circ$). Since the angles of a triangle sum to 180° , we know $A = D = 112^\circ$. But similarity alone is not congruence: two triangles can be similar without being congruent (one could be a scaled-up version of the other). To pin down congruence we need at least one pair of corresponding SIDES that are equal. That's exactly what choice C gives: BC and EF are corresponding sides (both opposite the 27° angle in their triangle). If I

× A) The measure of angle A

The measure of angle A adds no new info — we can already compute $A = 180^\circ - 27^\circ - 41^\circ = 112^\circ$ from the two given angles. Still no side information, so still can't confirm congruence.

× B) The length of side AB

AB is a side of triangle ABC, but this choice does not tell us anything about DEF's corresponding side. Only one triangle's side is described.

✓ C) The lengths of sides BC and EF

BC (opposite A in triangle ABC) and EF (opposite D in triangle DEF) are CORRESPONDING sides. Combined with the two equal angle pairs already known, this gives AAS congruence — sufficient to determine congruence.

× D) No additional information is necessary.

Two pairs of equal angles establish similarity, NOT congruence. The triangles could be different sizes. More information IS necessary.

Question 25 · Percent Increase (Solving Backwards)

Correct answer: D) 36

Worked solution

Increasing by 1,800% means adding 18 times the original to itself:

$$\text{new} = x + 18x = 19x.$$

$$19x = 684.$$

$$x = 684 / 19 = 36.$$

× A) 12,996

$12,996 = 684 \times 19$. This applies the increase forward instead of undoing it.

× B) 12,312

$12,312 = 684 \times 18$. Also applies the increase in the wrong direction.

× C) 38

$38 = 684 / 18$. Forgets to add the original x back — an increase of 1,800% multiplies by 19 total, not 18.

✓ D) 36

An 1,800% increase means the new amount is 19 times the original ($1x + 18x$). $19x = 684$, so $x = 36$.

Question 26 · Piecewise Linear Cost FunctionCorrect answer: A) $f(x) = 60x + 100$

Worked solution

The \$220 covers the first 2 hours. Beyond that, each extra hour costs some fixed rate r .**For 5 hours (3 hours beyond the base 2): total = $220 + 3r = 400 \rightarrow 3r = 180 \rightarrow r = \$60/\text{hour}$.****So for x hours ($x \geq 2$), extra hours = $(x - 2)$:**

$$f(x) = 220 + 60(x - 2)$$

$$= 220 + 60x - 120$$

$$= 60x + 100.$$

Check: $f(5) = 60(5) + 100 = 400 \checkmark$; $f(2) = 220 \checkmark$. $\checkmark$ A) $f(x) = 60x + 100$ **Additional hourly rate r satisfies $220 + 3r = 400$, so $r = 60$. Then $f(x) = 220 + 60(x - 2) = 60x + 100$.****Check: $f(2) = 220 \checkmark$, $f(5) = 400 \checkmark$.** $\times$ B) $f(x) = 60x + 220$ **$f(2) = 60(2) + 220 = 340 \neq 220$. Also $f(5) = 520 \neq 400$. Fails both anchor points.** $\times$ C) $f(x) = 80x$ **$f(2) = 160 \neq 220$. Doesn't respect the flat \$220 for the first two hours.** $\times$ D) $f(x) = 80x + 220$ **$f(2) = 380$ and $f(5) = 620$ — neither anchor point works.****Question 27 · Sum of Roots (Quadratic)**Correct answer: $29/3$

Worked solution

Expand both sides:

Left: $x^2 + x - 56$.

Right: $4x^2 - 28x$.

Bring everything to one side:

$$x^2 + x - 56 - 4x^2 + 28x = 0$$

$$-3x^2 + 29x - 56 = 0.$$

Multiply by -1 : $3x^2 - 29x + 56 = 0$.

For a quadratic $ax^2 + bx + c = 0$, sum of roots = $-b/a$.

Sum = $-(-29) / 3 = 29/3$.

Why this works

After expanding and collecting terms, the equation becomes $3x^2 - 29x + 56 = 0$. For a quadratic $ax^2 + bx + c = 0$, the sum of the solutions equals $-b/a$, so $-(-29)/3 = 29/3 (\approx 9.667)$.

Answer Key Summary

All 27 answers at a glance.

#	Skill	Answer
1	Reading a Line Graph	C
2	System of Linear Equations (Graphical)	A
3	Writing a Linear Inequality	D
4	Translating a Graph (Function Transformation)	A
5	Evaluating a Linear Function	D
6	Evaluating a Quadratic Function	77
7	Linear Inequality (Budget Constraint)	25
8	Two-Way Table Probability	C
9	Congruent Triangles (Angle Sum)	B
10	Scaling a Linear Equation	B
11	Exponent Rules (Product of Monomials)	B
12	Modeling with Functions (Linear vs Exponential)	B
13	Solving a Linear System	1
14	Function Transformation (Vertical Shift)	76
15	Interpreting Function Notation in Context	A
16	Exponential Growth Model (Doubling)	D
17	Exponential Function from a Table	D
18	Intercepts of a Linear Function	A
19	Linear Inequality — Table Check	A
20	Solving a Linear System	35
21	Pythagorean Theorem (Simplifying a Radical)	113
22	Scale Factor and Area	A
23	Equation of a Circle (Tangent to Axis)	C
24	Triangle Congruence (SSS/SAS/ASA/AAS)	C
25	Percent Increase (Solving Backwards)	D
26	Piecewise Linear Cost Function	A
27	Sum of Roots (Quadratic)	29/3

About SOMATH

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